

CSB24004-assignment8

```

import React, { useState } from "react";

export default function CourseEnrollmentDashboard() {
  const [studentsMap, setStudentsMap] = useState(new Map());
  const [name, setName] = useState("");
  const [gpa, setGpa] = useState("");
  const [coursesInput, setCoursesInput] = useState("");
  const [filterCourse, setFilterCourse] = useState("");

  const addStudent = () => {
    if (!name || !gpa) return;

    const newId = Date.now();
    const enrolledCourses = new Set(
      coursesInput
        .split(",")
        .map((c) => c.trim())
        .filter((c) => c.length > 0)
    );

    const newStudent = {
      id: newId,
      name,
      enrolledCourses,
      gpa: parseFloat(gpa),
    };

    const updatedMap = new Map(studentsMap);
    updatedMap.set(newId, newStudent);

    setStudentsMap(updatedMap);
    setName("");
    setGpa("");
    setCoursesInput("");
  };

  const removeStudent = (id) => {
    const updatedMap = new Map(studentsMap);
    updatedMap.delete(id);
    setStudentsMap(updatedMap);
  };

  const studentsArray = Array.from(studentsMap.values());

```

```
const sortedStudents = [...studentsArray].sort(
  (a, b) => b.gpa - a.gpa
);
```

```
const filteredStudents = filterCourse
  ? sortedStudents.filter((student) =>
    student.enrolledCourses.has(filterCourse)
  )
  : sortedStudents;
```

```
const n = studentsArray.length;
const filteringTimeComplexity = `O(n) where n = ${n}`;
```

```
const uniqueCourses = studentsArray.reduce((acc, student) => {
  student.enrolledCourses.forEach((course) => acc.add(course));
  return acc;
}, new Set());
```

```
return (
  <div className="p-6 max-w-4xl mx-auto">
    <h1 className="text-2xl font-bold mb-4">Course Enrollment Dashboard</h1>

    <div className="grid grid-cols-1 md:grid-cols-4 gap-3 mb-6">
      <input
        className="border p-2 rounded"
        placeholder="Student Name"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />
      <input
        className="border p-2 rounded"
        placeholder="GPA"
        type="number"
        step="0.01"
        value={gpa}
        onChange={(e) => setGpa(e.target.value)}
      />
      <input
        className="border p-2 rounded"
        placeholder="Courses (comma separated)"
        value={coursesInput}
        onChange={(e) => setCoursesInput(e.target.value)}
      />
      <button
```

```

        className="bg-blue-500 text-white rounded p-2"
        onClick={addStudent}
      >
        Add Student
      </button>
    </div>

    <div className="mb-4">
      <input
        className="border p-2 rounded"
        placeholder="Filter by Course"
        value={filterCourse}
        onChange={(e) => setFilterCourse(e.target.value)}
      />
    </div>

    <div className="mb-6">
      <h2 className="text-xl font-semibold mb-2">Students (Sorted by GPA)</h2>
      {filteredStudents.map((student) => (
        <div
          key={student.id}
          className="border rounded p-3 mb-2 flex justify-between items-center"
        >
          <div>
            <p className="font-medium">{student.name}</p>
            <p>GPA: {student.gpa}</p>
            <p>
              Courses: {Array.from(student.enrolledCourses).join(", ")}
            </p>
          </div>
          <button
            className="bg-red-500 text-white px-3 py-1 rounded"
            onClick={() => removeStudent(student.id)}
          >
            Remove
          </button>
        </div>
      ))}
      <div className="mt-4 p-3 bg-gray-100 rounded">
        <p className="font-semibold">Filtering Time Complexity:</p>
        <p>{filteringTimeComplexity}</p>
      </div>
    </div>

```

```

<div>
  <h2 className="text-xl font-semibold mb-2">All Unique Courses</h2>
  <ul className="list-disc pl-6">
    {Array.from(uniqueCourses).map((course, index) => (
      <li key={index}>{course}</li>
    ))}
  </ul>
</div>
</div>
);
}

```

Output

Course Enrollment Dashboard

Student Name: GPA: Courses (comma separated): Add Student

Filter by Course:

Students (Sorted by GPA)

joy
GPA: 8.5
Courses: physics, Chemestry
Remove

rocky
GPA: 7.5
Courses: physics, Chemistry, maths
Remove

Filtering Time Complexity:
 $O(n)$ where $n = 2$

All Unique Courses

- physics
- Chemestry
- Chemistry
- maths